Assignment 2

Digital Signal Analysis
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Due Date: 11:59pm on 1st April 2026

1. Solve all the questions and submit a handwritten document.
2. Plagiarism will be penalised.
3. You can use any coding/programming language to complete 'Coding' questions.
4. Submit a PDF of the form `<roll no>dsa2.pdf`.
5. The submission should be a zip folder named with your rollnumber-dsa2. Each coding question must be placed in a separate folder containing the corresponding code and results.

Question 1.

The Fourier transform of the signal

$$x(n) = \begin{cases} 1, & -M \leq n \leq M \\ 0, & \text{otherwise} \end{cases}$$

is given by

$$X(\omega) = 1 + 2\sum_{n=1}^{M} \cos(\omega n).$$

Show that the Fourier transforms of the signals

$$x_1(n) = \begin{cases} 1, & 0 \leq n \leq M \\ 0, & \text{otherwise} \end{cases} \quad \text{and} \quad x_2(n) = \begin{cases} 1, & -M \leq n \leq -1 \\ 0, & \text{otherwise} \end{cases}$$

are respectively given by

$$X_1(\omega) = \frac{1 - e^{-j\omega(M+1)}}{1 - e^{-j\omega}}, \tag{1}$$

$$X_2(\omega) = \frac{e^{j\omega} - e^{j\omega(M+1)}}{1 - e^{j\omega}}. \tag{2}$$

Hence, prove that

$$\begin{aligned} X(\omega) &= X_1(\omega) + X_2(\omega), && (3) \\ &= \frac{\sin\left((M+\frac{1}{2})\omega\right)}{\sin\left(\frac{\omega}{2}\right)}. && (4) \end{aligned}$$

Therefore, show that

$$1 + 2\sum_{n=1}^{M} \cos(\omega n) = \frac{\sin\left((M+\frac{1}{2})\omega\right)}{\sin\left(\frac{\omega}{2}\right)}. \tag{5}$$

Question 2.

Let $x(n)$ be an arbitrary signal (not necessarily real) with Fourier transform $X(\omega)$. Express the Fourier transforms of the following signals in terms of $X(\omega)$.

(a) $x^*(n)$

(b) $y(n) = x(n) - x(n-1)$

Question 3. (Coding)

Determine and sketch the Fourier transforms: $X_1(\omega)$, $X_2(\omega)$ and $X_3(\omega)$ of the following signals:

(a) $x_1(n) = \{1, 1, \underset{\uparrow}{1}, 1, 1\}$

(b) $x_2(n) = \{1, 0, 1, 0, \underset{\uparrow}{1}, 0, 1, 0, 1\}$

(c) $x_3(n) = \{1, 0, 0, 1, 0, 0, \underset{\uparrow}{1}, 0, 0, 1, 0, 0, 1\}$

(d) Is there any relation between $X_1(\omega), X_2(\omega)$and$X_3(\omega)$? What is its physical meaning?

Question 4.

Given the 8-point DFT of the sequence:

$$x(n) = \begin{cases} 1, & 0 \le n \le 3 \\ 0, & 4 \le n \le 7 \end{cases}$$

Compute the DFT of the sequences:

(a)

$$x_1(n) = \begin{cases} 1, & n = 0 \\ 0, & 1 \le n \le 4 \\ 1, & 5 \le n \le 7 \end{cases}$$

(b)

$$x_2(n) = \begin{cases} 0, & 0 \le n \le 1 \\ 1, & 2 \le n \le 5 \\ 0, & 6 \le n \le 7 \end{cases}$$

Question 5. (Coding)

Consider a finite-duration sequence

$$x(n) = \{\underset{\uparrow}{0}, 1, 2, 3, 4\}$$

(a) Sketch the sequence $s(n)$ with 6-point DFT:

$$S(k) = W_2^* X(k), \quad k = 0, 1, \ldots, 6$$

(b) Determine the sequence $y(n)$ with 6-point DFT:

$$Y(k) = \mathrm{Re}\{X(k)\}$$

(c) Determine the sequence $v(n)$ with 6-point DFT :

$$V(k) = \mathrm{Im}\{X(k)\}$$

Question 6.

Determine the eight-point DFT of the signal:

$$x(n) = \{1, 1, 1, 1, 1, 1, 0, 0\}$$

and sketch it's magnitude and phase.

Question 7.

Given:

$$x(n) = \{1, 2, \underset{\uparrow}{3}, 2, 1, 0\}$$

$$v(n) = \{3, 2, 1, 0, 1, 2\}$$

Is there any relation between Fourier transform $X(\omega)$ of $x(n)$ signal and 6-point DFT $V(k)$ of $v(n)$ signal? Explain.

Question 8. (Coding)

The signal:

$$x(n) = a^{|n|}, \quad -1 < a < 1$$

has a Fourier Transform

$$X(\omega) = \frac{1 - a^2}{1 - 2a\cos\omega + a^2}$$

(a) Plot $X(\omega)$ for $0 \leq \omega \leq 2\pi$, $a = 0.8$

(b) Reconstruct and plot $X(\omega)$ from its samples:

$$X\left(\frac{2\pi k}{N}\right), \quad 0 \leq k \leq N - 1$$

a. For $N = 20$

b. For $N = 100$

(c) Compare the spectra obtained in parts a. and b. in (b) with the original spectrum $X(\omega)$ and explain the differences.

(d) Illustrate the time-domain aliasing for $N = 20$